

HW 1

① Gauss's Law for Electricity

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{enc}}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

- The net electric flux through a closed surface is related to the total charge enclosed.
- There exists a diverging $\vec{E}$ field if there exists non-zero charge density.

Gauss's Law for Magnetism

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

- Any closed surface will have zero magnetic flux.
- There doesn't exist a diverging $\vec{B}$ field because the magnetic monopole does not exist.

Faraday's Law

$$\oint_P \vec{E} \cdot d\vec{\ell} = - \frac{d\Phi_B}{dt}$$

- Relates the circulation of the $\vec{E}$ field around a closed path to the changing magnetic flux through the surface bounded by the loop.

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

- A changing $\vec{B}$ field will induce a curling $\vec{E}$ field.

Ampere-Maxwell Law

$$\oint_P \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

- Relates the circulation of the $\vec{B}$ field around a closed path to the enclosed current and changing electric flux through the surface bounded by the loop.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

- A changing $\vec{E}$ field or a non-zero current density will induce a curling $\vec{B}$ field.

② Both

$$\vec{\nabla} \cdot \vec{E} = 0 \quad \text{and} \quad \vec{\nabla} \cdot \vec{B} = 0 \quad \text{in a vacuum because}$$

there exists no charge and $\vec{\nabla} \cdot \vec{B}$ is always zero.

$$\begin{aligned}
 \textcircled{3} \quad \vec{\nabla} \times (\vec{\nabla} \times \vec{B}) &= \vec{\nabla} (\vec{\nabla} \cdot \vec{B}) - \vec{\nabla}^2 \vec{B} \\
 \Rightarrow \vec{\nabla} \times \left(\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= -\vec{\nabla}^2 \vec{B} \\
 \Rightarrow \mu_0 (\vec{\nabla} \times \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{E}) &= -\vec{\nabla}^2 \vec{B} \\
 \hookrightarrow \vec{J} &= 0 \text{ in a vacuum} \\
 \Rightarrow \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(-\frac{\partial \vec{B}}{\partial t} \right) &= -\vec{\nabla}^2 \vec{B} \\
 \Rightarrow \boxed{\mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = \vec{\nabla}^2 \vec{B}}
 \end{aligned}$$

④ Thus we can conclude that $\vec{E}$ and $\vec{B}$ fields exist as light waves propagating through vacuum.

It has the properties:

- Do not require a medium to propagate
- $\vec{E} \times \vec{B}$ points in the direction of propagation
- $\vec{E}$ is always perpendicular to $\vec{B}$
- It has speed

$$V = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

- $\vec{E}$ and $\vec{B}$ oscillate so they perpetually induce each other.